

Original Article

Antifungal Triazole Derivative Triadimefon Induces Ectopic Maxillary Cartilage by Altering the Morphogenesis of the First Branchial Arch

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BACKGROUND: The triazole derivative, triadimefon (FON), induces branchial arch abnormalities in post-implantation rat embryos cultured in vitro, and cranio-facial malformations in mouse fetuses. Ectopic maxillary cartilage has been also described as a typical FON-related malformation. This work studies the morphogenesis of the ectopic cartilage in rat embryos and fetuses exposed in vivo to FON during the early postimplantation period. **METHODS:** Pregnant rats were treated with 0, 250, and 500 mg/kg FON on Day 9.5 of pregnancy (D9.5) and sacrificed at term (D20), during the early fetal period (D17) or at different embryogenetic periods (D10, D11, D12). The skeleton was examined after stain of bone and cartilage or of cartilage alone respectively at term or at D17. The neural crest cell (NCC) migration and compaction was investigated at D10 and D11 and the cranial nerve organization described at D12. **RESULTS:** Triadimefon is teratogenic in rats under the chosen experimental conditions. The malformations were at the level of the cranio-facial and axial skeleton at term and of the hindbrain nerves in embryos. A NCC abnormal migration and compaction was observed at the level of the first branchial arch: in FON-exposed embryos NCC were detected at the level of both maxillary and mandibular processes, whereas control embryos showed the immunostained tissue only at the level of the mandibular bud. **CONCLUSIONS:** The pathogenic pathway proposed to explain the ectopic cartilage is the displacement of part of the NCC-derived tissues at the maxillary region of the first branchial arch. *Birth Defects Res (Part B)* 80:2–11, 2007. © 2006 Wiley-Liss, Inc.

Key words: triazole; triadimefon; teratogenesis; retinoic acid; cranio-facial malformations; axial malformations; branchial arch; maxillary bud; neural crest cell

INTRODUCTION

Antifungal triazole-derivatives are teratogenic agents, believed to act by inhibiting the embryonic catabolism of the morphogen retinoic acid (RA) (for an extensive review see Menegola et al., 2006).

The metabolite of vitamin A, RA acts as a morphogen in vertebrates (Ross et al., 2000). A direct relationship between RA concentration, Hox gene expression, cranio-facial (Osumi-Yamashita, 1996), vertebral (Kessel and Gruss, 1991; Kawanishi et al., 2003) or limb (Niederreither et al., 2002) specification has been documented.

Craniofacial bones in vertebrates with jaws (gnathostomes) originate entirely from the frontal and branchial arch ectomesenchyme, a mesenchymal tissue formed by the condensation of cells (neural crest cells [NCC], a migratory and multipotent cell population unique to vertebrates) migrated from the brain (Hall, 1999; Le Douarin and Kalcheim, 1999; Le Douarin and Dupin, 2003; Santagati and Rijli, 2003) (Fig. 1). Specifically, in all vertebrates cranial NCC migrate in a conserved pattern of three streams: the mandibular (or trigeminal) stream,

emerging from the mesencephalon and from the hind-brain rhombomere (r) 2, migrates to the anterior head and to the first branchial arch; the hyoid stream, directed from r4 to the second branchial arch; and the common branchial stream, directed from r6 to the branchial arches III–VI (Lumsden et al., 1991; Epperlein et al., 2000; Meulemans and Bronner-Fraser, 2002; Trainor et al., 2002) (Fig. 1).

Neural crest cells that migrate to the first branchial arch form a dorsal (maxillary) and a ventral (mandibular) ectomesenchymal condensation (Francis-West et al., 1998; Chai et al., 2000; Meulemans and Bronner-Fraser, 2002). Recently, the assumption that the embryonic cartilaginous skeleton of the upper jaw (palatoquadrate cartilage) is formed from the maxillary component and that the embryonic lower jaw skeleton (Meckel's

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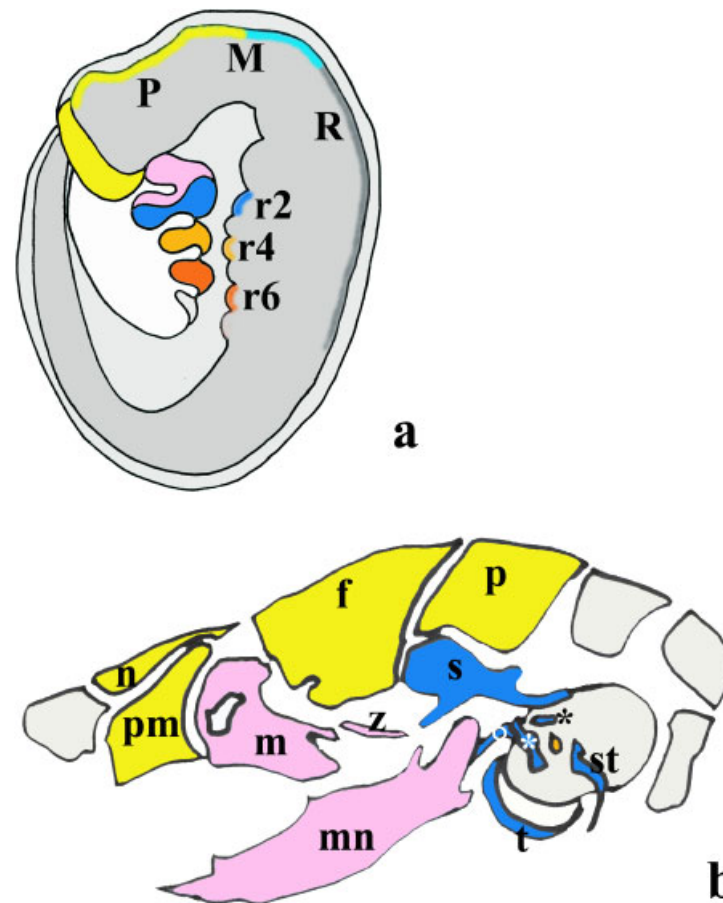


Fig. 1. Morphogenesis of the cranial structures derived from the encephalic NCC. **A:** During the early embryonic development, NCC arising from the encephalon migrate to the mesenchymal layers of the frontonasal region of the head and of branchial arches. In particular, from prosencephalon (P) and part of mesencephalon (M) NCC migrate anteriorly the optic vesicle (yellow area), from the hindbrain (rhombencephalon, R) NCC migrate to the branchial apparatus. In particular from rhombomere 2 (r2) NCC reach the mandibular part of the first branchial arch, whereas from r4 and r6 NCC migrate to colonize respectively the second and third branchial arch. Note that the maxillary part of the first branchial arch is not described as a NCC migration. **B:** Scheme illustrating the origin of cranial bones. Yellow: skeletal elements derived from the frontonasal ectomesenchyme (n, nasal; f, frontal; p, parietal; pm, premaxillary bones). Pink: bones derived from the mesenchyme of the maxillary bud of the first branchial arch (m, maxilla; z, zygomatic bone; mn, mandible). Blue: elements derived from the mandibular bud of the first branchial arch (s, squamosal bone; t, tympanic ring; st, styloid process; o, Meckel's cartilage; white *, malleus; black *, incus). Orange: the stapes is derived from the second branchial arch ectomesenchyme.

Table 1
Immunostaining Procedures

Stained Structures	Staining Method	Fixative	Primary Antibody	Secondary Antibody	Visualization
NCC	Whole mount	Dent's (DMSO: methanol, 1:4) Overnight -20°C	Mouse anti-CRABP PBS, 1:250 3 days $+4^{\circ}\text{C}$	Goat anti-mouse IgG-peroxidase PBS, 1:40 Overnight $+4^{\circ}\text{C}$	4-Cl-naphtol + oxygen peroxide
NCC	Immuno- histochemistry	Dent's (DMSO: methanol, 1:4) Overnight -20°C	Mouse anti-CRABP PBS, 1:250 Overnight $+4^{\circ}\text{C}$	Goat anti-mouse IgG-peroxidase PBS, 1:40 1 hr RT	DAB + oxygen peroxide
Nerves and ganglia	Whole mount	Paraformaldehyde 6 hr $+4^{\circ}\text{C}$	Mouse anti-neuro filament 160 kDa PBS, 1:100 3 days $+4^{\circ}\text{C}$	Goat anti-mouse IgG-fluorescein PBS, 1:50 Overnight $+4^{\circ}\text{C}$	Under a fluorescent stereomicroscope

NCC=neural crest cells; DMSO=dimethyl sulfoxide; CRABP=cellular retinoic acid binding protein; PBS=phosphate saline buffer; RT=room temperature; DAB=diaminobenzidine.

cartilage) is formed from the mandibular component of the first branchial arch has been rejected because, after staining the two components with molecular markers, Cerny et al. (2004) showed that both palatoquadrate and Meckel's cartilage derive from mandibular condensation, whereas the maxillary component gives rise to cartilaginous anterior neurocranial structures. Neural crest cells migrating into the first branchial arch contribute exclusively to mandibular prominence, whereas the major contribution to the maxillary bud is from the post-optic mesenchyme (Lee et al., 2004).

In the post-optic mesenchyme a local source of RA initiates a regulatory cascade, controlling cell proliferation, apoptosis, and extracellular matrix synthesis that coordinate forebrain and frontonasal morphogenesis. The derivatives of maxillary prominence in chickens are dermal bones such as maxillary, palatine, and jugal bones (Lee et al., 2004).

Among agents that induce craniofacial abnormalities, triazole derivatives are described as molecules able to

Table 3
External Abnormalities Recorded at Term of Pregnancy^a

	Control	FON (250 mg/kg)	FON (500 mg/kg)
No. of examined fetus (Malformed fetuses/live fetuses) 100 ^b	165	100 54.0 ^c	125 85.6 ^{c,d}
Ablepharia		5.0	24.0
Ear duplication			8.0
Ectopic ear			6.4
Micrognathia			7.2
Microstomia		54.0	65.6
Microtia			16.8
Cleft palate		2.0	16.8
Tongue protrusion			25.6

^aD20 (%).

^bStatistical analysis carried out.

^c $P < 0.01$ vs. control.

^d $P < 0.01$ vs. FON 250.

Table 2
Litter Parameters Recorded at Term of Pregnancy (D20)

	Control	FON (250 mg/kg)	FON (500 mg/kg)
No. of treated females	15	15	15
Implantations (means $\pm$ SD)	13.27 $\pm$ 2.76	13.00 $\pm$ 3.38	14.56 $\pm$ 2.35
Postimplantation loss index (%) ^a	3.42	3.85	4.58
Fetal weight (mean $\pm$ SD)	4.08 $\pm$ 0.18	3.83 $\pm$ 0.27	4.01 $\pm$ 0.27
Placental weight (mean $\pm$ SD)	0.55 $\pm$ 0.07	0.50 $\pm$ 0.07	0.50 $\pm$ 0.06

Fig. 2. Skeletal appearance of cranial and axial structure after double staining for bone (red) and cartilage (blue) in D20 controls (A–C, J, L, N) and FON-treated (D–I, K, M, O) fetuses. **A:** Lateral appearance of the normal rat cranium at term of gestation. The maxillary structures are all dermally derived (+, premaxillary; ○, maxillary; >, zygomatic; black *, squamosal bones), so no cartilaginous elements are visible. At the mandible level, the dental bone (#) and Meckel's cartilage (M) are recognizable. The tympanic ring (t, osseus) and the cartilaginous styloid process (st) are clearly visible near the otic capsule (O). **B:** Enlargement of (A). Note the normal structure of the squamosal bone (black *) with the retro tympanic (§) and zygomatic (white *) processes. Meckel's cartilage (M) reaches the ear ossicles (m, malleus; l, incus; s, stapes). ○, Zygomatic process of the maxillary bone; #, dental bone; >, zygomatic bone. **C:** View of the base of the cranium of a control fetus. B, basioccipital bone; b and +, basisphenoid and alisphenoid bones; ○, pterygoid process; @, palatine; black * and white *, basal portion of squamosal and maxillary bones; O, otic capsule; t, tympanic ring. **D,E:** Lateral view of a FON 250 fetus. Note the agenesis of the styloid process, the reduction of the tympanic ring (X), the abnormal shape of the squamosal (black *) and zygomatic (>) bones whereas the premaxilla (+) and maxilla (○) appear normally structured. In the enlargement the ectopic cartilage at the level of the upper jaw is visible (>>) and Meckel's cartilage, abnormal, fused with the ear ossicles (>>>). **F:** Examination of the base of the skull in a FON 250 fetus shows a partial cleft palate (>), the abnormal position and shape of alisphenoid bones (+), in reference to the basisphenoid (b), the abnormal shape of pterygoid processes (○), the upper jaw ectopic cartilage (>>) joining the basal cartilages and the abnormal Meckel's cartilage reaching the middle ear fused to the ear ossicles (>>>). **G,H:** Lateral view of a FON 500 fetus. Note the agenesis of styloid process and tympanic ring, the fusion between the ear ossicles (>>>), the large ectopic cartilage (>>) joining the upper (++) and lower (#) jaw. **I:** Base of the skull of a FON 500 fetus. Note the large cleft palate (>), the abnormal position and shape of alisphenoid bones (+), in reference to the basisphenoid, the abnormal pterygoid process. Note the upper jaw ectopic cartilage (>>) joining the basal cartilages and the abnormal Meckel's cartilage reaching the middle ear fused to the ear ossicles (>>>). **J:** Isolated sternum of a control fetus. Seven ribs (I–VII) reach the sternum inducing six (1–6) ossification centers (sternebrae). The eighth rib, as well as the other posterior ribs, is floating (+). **K:** Isolated sternum of a FON-treated fetus: the first and second right ribs are inserted together, inducing a monolateral shift in the insertion of the posterior ribs (*, eighth is inserted into the sternum). At the left side of the sternum it is possible to observe a normal insertion of ribs I–VII, but the eighth appears approached to the sternum (○). The asymmetric insertion of ribs into the sternum induces the formation of an extra sternebra site (#). **L:** Lateral view of the cervical and thoracic region of a control fetus. There are seven cervical vertebrae. The first two (atlas [A] and axis [Ax]) show characteristic morphology. The sixth is characterized by the presence of *tubercula anteriora* (>). The thoracic segments, also characterized by the presence of the ribs, are 13 (I–XIII). The morphology of the neural spine is typical of thoracic vertebrae: I–IX, caudally directed; XI–XIII, cranially directed; and the tenth with a hammer-like shape (○). **M:** Lateral view of the cervical and thoracic region of a FON-treated fetus. The hammer-like shape of the neural spine is shifted to the thoracic vertebra XI (○). **N:** Ventral view of the lumbar and sacral region of a control fetus. The lumbar vertebrae are six (1–6), all characterized by the presence of cranially directed transverse processes (>). The sacral region is characterized by five vertebrae (I–V) showing transverse processes enlarged (>>) and directed toward the ileum (*). The fifth vertebra is localized at the same level of the pubis (+). **O:** Ventral view of the lumbar and sacral region of a FON-treated fetus. An anterior homeotic respecification is easily detectable: lumbar ribs (○) are inserted on first lumbar vertebra bilaterally, whereas the first sacral vertebra shows a monolateral (left) respecification of the transverse process morphology (>).

Statistical analysis carried out on all parameters.

^aCalculated as (total implantation–live fetuses/total implantations) $\times$ 100.

induce abnormalities similar to those observed after RA exposure both in vivo and in vitro. In particular, the in vitro exposure of postimplantation rat embryos to a number of triazole derivatives induces specific malformations at the branchial apparatus (fusion between the first and second branchial arch, hypoplastic branchial arch I or II), at the associated cranial nerves (V, VII, IX, X, fused, and disorganized), related to the incorrect migration of NCC streams from hindbrain to the branchial arches and to the abnormal segmentation of the rhombencephalic vesicle (Tiboni, 1993; Menegola et al., 2000, 2001a, 2003, 2004, 2005a). The treatment of mouse embryos with two triazole antifungals (triadimefon [FON] and fluconazole) during the early embryonic period, induced craniofacial (Menegola et al., 2005b; Tiboni and Giampietro, 2005) and axial abnormalities (Menegola et al., 2005b). Interestingly, maxillary ectopic cartilage, similar to that observed by Morriss-Kay (1993) after RA exposure, was detected after in vivo

exposure of mouse embryos to FON (Menegola et al., 2005a).

The present work studies the genesis of ectopic cartilage in rat embryos and fetuses exposed in vivo during the early embryonic period to the triazole derivative FON. The rat was chosen because of the more synchronous development of embryos in this species in comparison to mouse.

MATERIALS AND METHODS

CD:CrI rats (Charles River, Calco, Italy), housed in a thermostatically maintained room ($T = 22 \pm 2^\circ\text{C}$, relative humidity = $55 \pm 5\%$) with a 12-hr light cycle (light from 6:00 a.m. to 6:00 p.m.), with food (Italiana Mangimi, Settimo Milanese, Italy) and tap water ad libitum, were used for this study. Virgin females were mated overnight with males of proven fertility. The day of positive vaginal smear was considered as Day 0 of gestation (D0).

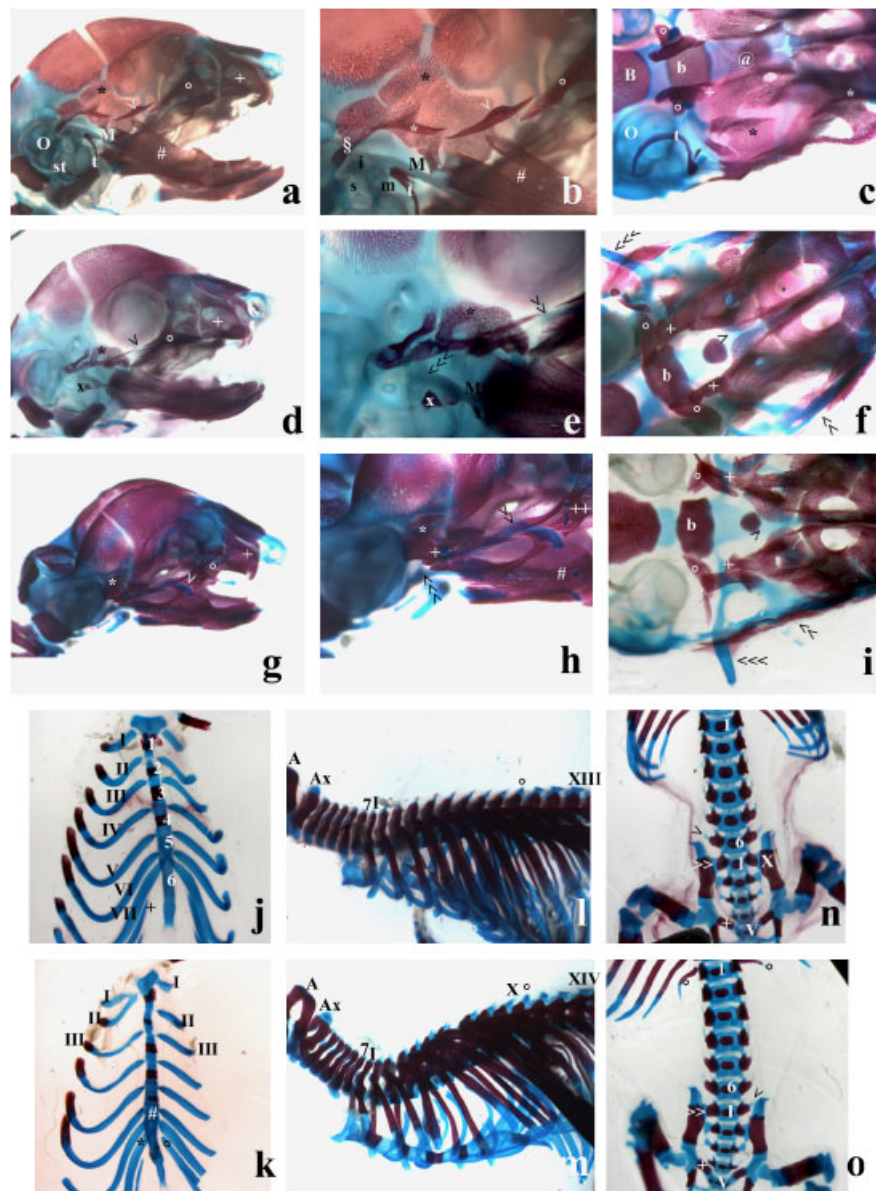


Table 4
Craniofacial Skeletal Abnormalities Recorded at Term of Pregnancy^a

	Control	FON (250 mg/kg)	FON (500 mg/kg)
No. of examined fetuses	165	100	125
(Malformed fetuses/live fetuses) × 100 ^b		100 ^c	97.6 ^c
Reduced cranium ossification ^b		5.0	9.6
Skull basis ^b		39.0 ^c	72.8 ^{c,d}
Hyoid			
Abnormal shape		4.0	2.4
Unossified		14.0	31.2
Alisphenoid (abnormal and ectopic)		25.0	48.0
Basisphenoid (reduction/agenesis)		2.0	
Ectopic cartilage between maxilla-pterygoid			25.6
Tympanic ring ^b		31.0 ^c	55.2 ^{c,e}
Agenesis		12.0	35.2
Reduction/abnormal shape		19.0	20.0
Maxilla, zygomatic, squamosal bones ^b		90.0 ^c	81.6 ^c
Maxilla (abnormal shape)		87.0	80.0
Squamosal bone (abnormal shape)		82.0	66.4
Zygomatic			
Abnormal shape		68.0	28.8
Agenesis		14.0	40.0
Ectopic cartilage			
Rudimental		71.0	29.6
Compact and linear		10.0	36.8
Mandible ^b		78.0 ^c	97.6 ^{c,d}
Short		78.0	97.6
Abnormal shape		10.0	40.0
Abnormal Meckel's cartilage		54.0	52.8
Fused with zygomatic		1.0	
Ectopic cartilage			
Rudimental		10.0	40.0
Joining mandible and maxilla		1.0	37.6
Middle ear ossicles ^b		54.0 ^c	52.8 ^c
Fused		54.0	52.8

^aD20 (%).

^bStatistical analysis carried out.

^c $P < 0.01$ vs. control.

^d $P < 0.01$ vs. FON 250.

^e $P < 0.05$ vs. FON 250.

Table 5
Axial Skeletal Malformations Recorded at Term of Pregnancy^a

	Control	FON (250 mg/kg)	FON (500 mg/kg)
No. of examined fetuses	165	99	125
(Malformed fetuses/live fetuses) × 100 ^b	6.1	65.6 ^c	67.2 ^c
Thoracic region ^b		16.2 ^c	21.6 ^c
Duplications		15.2	21.6
Wavy ribs		1.0	
Sternum ^b		12.1 ^c	8.0 ^d
1st and 2nd ribs reaching manubrium			0.8
8th rib approaching sternum		9.1	6.4
7th rib approaching sternum		3.0	
Asymmetrical sternbrae			1.6
Lumbar region ^b	6.1	47.5 ^c	44.0 ^c
Lumbar ribs	6.1	47.5	44.0
Sacral region ^b		3.0 ^c	15.2 ^{c,e}
1st vertebra like lumbar ones		3.0	14.4
2nd and 3rd vertebrae like 1st and 2nd ones			4.8
Extra hemivertebra (3rd vertebra)			0.8

^aD20 (%).

^bStatistical analysis carried out.

^c $P < 0.01$ vs. control.

^d $P < 0.05$ vs. control.

^e $P < 0.05$ vs. FON 250.

Pregnant females were treated at D9.5 (5:00 p.m.) with 0, 250, and 500 mg/kg body weight FON (purity = 96.1%; gift of Bayer, Italy), dissolved in acetone diluted 1:9 in corn oil. The final volume of treatment was 5 ml/kg body weight. The controls were treated with the same volume of vehicle. Females were checked for signs of toxicity after the treatment until the sacrifice and were weighted at the start of gestation, at treatment time and at times of sacrifice. Some females from control and FON 250 and 500 groups were sacrificed during the embryogenetic period, at D10 (1:00 p.m.), D11 (10:00 a.m.), and D12 (10:00 a.m.). After explantation, embryos were morphologically examined and, as an indicator of developmental degree, the somite number was recorded. D10 and D11 embryos were processed for the whole mount NCC staining by using antibodies anti-CRABP (cellular retinoic acid binding protein) marker for NCC as described by Wei et al. (1999) and Menegola et al. (2003). D12 embryos were used for the whole mount immunostaining of cranial nerves and ganglia, according to the method proposed by Mark et al. (1993). Immunostaining procedures are summarized in Table 1.

Some females of control and FON 500 groups were sacrificed during the fetal period (D17) for the staining of cartilage. Females from all groups were sacrificed at term of gestation (D20), when the litter weight, number of implantations, resorptions, live, and dead fetuses were recorded. The postimplantation loss index [(total implantation–live fetuses/total implantation) $\times$ 100] was calculated. At D20, after external examination, live fetuses were weighed and all processed for skeletal double staining of bone and cartilage. The nomenclature used for identify skeletal elements was in accordance to Depew et al. (2002).

Quantitative parameters, expressed as media $\pm$ standard deviation (SD), were analyzed by using the statistical test ANOVA followed by Tukey's test. Qualitative parameters were expressed as percentage and statistically analyzed by using χ^2 test. The level of significance was set at $P < 0.05$.

Staining of Cartilaginous Elements in D17 Fetuses

After sacrifice, fetuses were dehydrated overnight in ethanol 96% and maintained in the staining solution at pH 2.8 (Alcian blue GX [0.14% dissolved in ethanol 70%] [Sigma, St. Louis, MO]:acetic acid:ethanol 70% [5:8:50]) for a few days until staining, in 0.2% KOH until maceration of soft tissues, and then in clearing solution (ethanol 70%:glycerin:benzyl alcohol [2:2:1]). Samples were conserved in ethanol 70%:glycerin (1:1).

Double Staining of Bone and Cartilage in D20 Fetuses

The double staining of bone and cartilage was carried out according to the method of Kimmel and Trammel (1981), partially modified as described previously (Menegola et al., 2001b).

RESULTS

Results at Term of Gestation (D20)

To verify the teratogenic potential of FON on rat skeletons, 15 females for each group were treated on

D9.5 and sacrificed at term of gestation (D20). No signs of maternal toxicity was detected in the treated group, as shown by clinical signs and by the weight gain after treatment (data not shown). Implantation number, postimplantation loss index, and fetal and placental weight were similar to control for both treated groups (Table 2). Fetuses explanted from dams dosed with FON showed, after external examination, abnormalities at the level of the head. The malformations, dose-related, included microstomia and cleft palate (Table 3). Craniofacial-specific and severe abnormalities were observed after skeleton staining in most of the fetuses explanted from FON-treated mothers (Table 4, Fig. 2D–I). The abnormal structures were the upper and lower jaws, the tympanic ring, and the base of the skull. The abnormalities were abnormal shape, reduction, agenesis, and fusion of skeletal elements. Ectopic cartilage at the level of upper jaw was observed in both treated groups, visible as ectopic cartilaginous fragments or as a large element. Fetuses (1% and 38%, respectively) of group FON 250 and FON 500 was characterized by a large shape cartilage joining the upper and lower jaw together (Table 4, Fig. 2D,E,G,H). The presence of ectopic cartilage was always coupled with ectopic cartilaginous elements at the base of the skull. These elements were usually fused with the ectopic cartilage (Fig. 2F,I).

As far as the axial elements are concerned (vertebrae, ribs, sternum), the recorded abnormalities were duplications and anterior homeotic transformations (Table 5, Fig. 2K,M,O).

Results at D17

To analyze the ectopic cartilage, D17 fetuses were explanted from five females of control and FON 500 groups. The morphology of chondrocranium in controls is shown in Figure 3A,C,E,G. FON 500 fetuses were characterized by a shorter and abnormal Meckel's cartilage (Fig. 3B,D,F), by a large cartilaginous element in the maxillary region (Fig. 3B,D), and by the abnormal shape of the cartilaginous elements at the base of the skull (Fig. 3H).

Results at Midgestation

To verify the development of control embryos and FON groups, NCC migration, and cranial nerve organization, five females for each group and each day were treated on D9.5 and sacrificed at D10, D11, D12. No differences were recorded as far as the morphology and the somite number are concerned (data not shown). D10 and D11 embryos were immunostained with anti-CRABP antibodies to visualize the NCC. D10 control embryos were characterized by NCC migrating from the encephalon to the fronto-nasal and branchial region (Fig. 4A). The NCC migration at the cephalic region was completed at D11, and the ectomesenchyme was compacted at the level of the fronto-nasal region and, separately, at the level of each branchial arch (Fig. 4D). D10 FON 250 and FON 500 embryos were characterized by a progressive disorganization of the migration streams (Fig. 4B,C), resulting in a diffuse immunostained mass in the fronto-nasal and branchial region at D10 (Fig. 4B,C) and in an abnormally compacted ectomesenchyme at D11 (Fig. 4E,F). The immunostaining of cranial nerves at D12

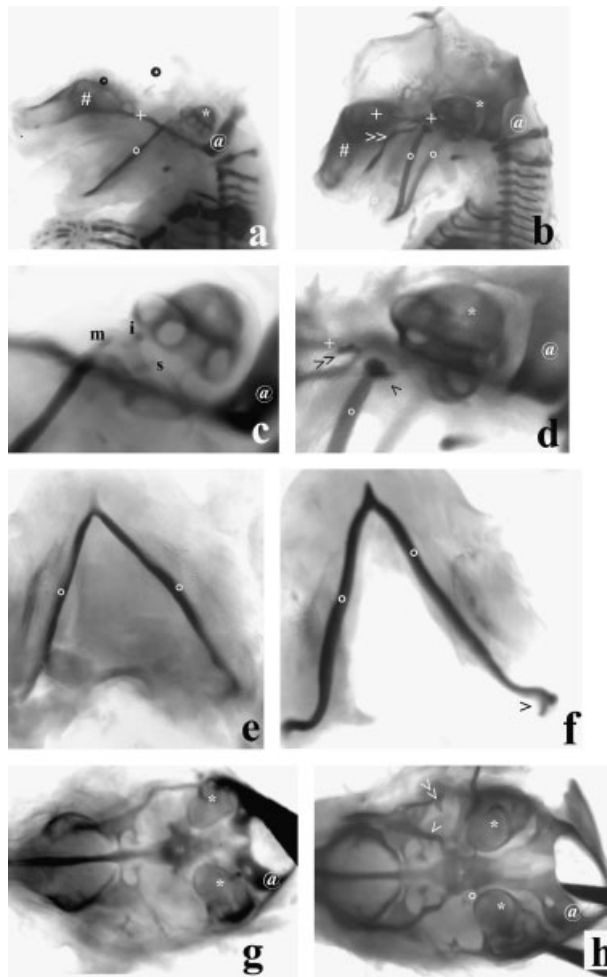


Fig. 3. Appearance of cranial structures after staining of cartilage in D17 controls (A,C,E,G) and FON-500 treated (B,D,F,H) fetuses. A,C: Lateral view of the head of a D17 control fetus. The nasal (#) and otic (*) capsules, the base of the skull (+) and the occipital (@) cartilages are clearly detectable. No cartilaginous elements are visible at the maxillary level. On the contrary, Meckel's cartilage in the mandible (o) at this developmental time is well-organized and reaches the middle ear (where the ear ossicles malleus (m), incus (i), and stapes (s) are visible). B,D: Lateral view of a FON-treated fetus. All the elements of the neurocranium are visible (#, nasal capsule; *, otic capsule; @, occipital cartilage; +, basal cartilages. Left and right Meckel's cartilages (o) are present but with an abnormal shape (>). They are unable to reach the middle ear (where the ear ossicles are not visible). In addition, an extra cartilaginous element is visible at the maxillary level (>>), appearing as a linear structure fused proximally to the base of the skull. E: Ventral view of the mandible of a control fetus. Right and left Meckel's cartilages are visible. Note the linear structure of this skeletal element. F: Ventral view of the mandible of a FON fetus. Note the proximal abnormal shape (>). G: Ventral view of the base of the skull of a control fetus. The occipital cartilage (@), the otic (*), and nasal (#) capsules join the cartilaginous basal plate of the skull. Note the typical shape of the different skeletal elements. H: Ventral view of the base of the skull of a FON fetus. The nasal (#) and otic (*) capsules as well as the occipital cartilages appear normally developed. The maxillary ectopic cartilage is visible (>>), fused with the basal cartilages that appear altered morphologically (>).

showed in control embryos, the well-defined and distinct nerves V, VII, VIII, IX, and X (Fig. 4G), which were fused together in embryos exposed to FON, with a more severe pattern of fusion in FON 500 group (Fig. 4H,I).

DISCUSSION

The data obtained in the present study show that FON is teratogenic in rats dosed at D9.5. The target structures are those controlled by the morphogen RA and the abnormalities observed are similar to those due

to exogenous RA exposure [cleft palate (Young et al., 2000), cranio-facial malformations (Osumi-Yamashita, 1996), homeotic respecification of vertebral identity (Kessel and Gruss, 1991; Kawanishi et al., 2003)] Moreover, the FON-related induction of maxillary ectopic cartilage was confirmed in the present study. The ectopic cartilage observed on maxilla was a cartilaginous structure visible clearly at D17 as well. After FON 500 treatment the morphology of the ectopic cartilage was linear, parallel to the mandibular Meckel's cartilage, fused with the basal cartilages (appearing in FON fetuses with an abnormal morphology). In the worst cases, the ectopic cartilage was a large element joining the



Fig. 4. Control (A,D,G) and FON-treated (B,C,E,F,H,I) embryos explanted at D10 (A–C), D11 (D–F), and D12 (G–I) and immunostained with antibodies anti-CRABP (A–F) or anti-neurofilament 160 kDa (G–I). A–C: Visualization at D10 of NCC migration in control embryos (A) and in FON 250 (B) or FON 500 (C). In controls (A) NCC appear as a migrating cell population. Three different streams, starting at the encephalon (*) and reaching the branchial apparatus (I, II, first and second branchial arches), are clearly visible (>, >>, >>>) and an additional stream (++) reaching the frontonasal region (o) is detectable. In FON 250 embryos (B) NCC migrating from the encephalon to the first branchial arch (>) appear mixed together with those migrating into the frontonasal region (++) forming a unique indistinguishable mass (#). This effect was more marked in FON 500 embryos, where a confluent unique migrating stream (>, +) was observed and a diffuse immunostained region was detected at the branchial and frontonasal region. D–F: At D11 in controls (D) a compacted ectomesenchyme (NCC-derived tissue forming a compacted immunostained mass specifically at the level of restricted areas) is visible at the level of the frontonasal region (o) and at the level of the branchial apparatus (I, II, first and second branchial arches), whereas the maxillary process (+) appeared only lightly stained. In FON 250 (E) and 500 (F) embryos, the immunostained tissue was restricted to the cephalic region of the first branchial arch (o) and to the proximal region of the second branchial arch (*) and an immunostained mass is observable in the region of the maxillary process (+). These effects seem to be dose-related. G–I: Fluorescent visualization of cranial nerve structure in controls at D12. The branchial nerves (#, trigeminal; o, facial; +, glossopharyngeal; \$, vagus) and the vestibulo-cochlear (*) nerve are clearly detectable. Note the ophthalmic, maxillary, and mandibular branches of the trigeminal nerve. The analysis on FON embryos after immunostaining showed fusion (>) between the cranial nerves at both dose levels. This effect was clearly dose-related.

maxillary and mandibular bones; in these cases a severe microstomia was detectable upon external analysis.

During normal development, the maxillary component of the first branchial arch gives rise to the dermal bones of the maxilla and to the cartilaginous elements at the base of the skull. By contrast, palatoquadrate and Meckel's cartilage (usually considered maxillary and mandibular cartilages) originate exclusively from the mandibular prominence (Cerny et al., 2004). Regarding the role of mesenchyme in organizing the maxillary and mandibular buds, evidence shows that the ectomesenchyme derived from encephalic NCC plays a role only in the formation of the mandibular part of the first branchial arch, whereas the maxillary component exclusively originates from the post-optic mesoderm (Lee et al., 2004).

Neural crest cell immunostaining at D10 and D11 of FON-treated embryos indicates clearly a dose-related alteration of NCC migration from the encephalon to frontonasal and first branchial arch regions: migrating NCC appear at D10 mixing and migrating in a unique stream, with abnormal ectomesenchymal condensation originating at D11. In particular, at D11 the ectopic staining of the maxillary prominence is evident if compared with the controls. This observation leads to the hypothesis that an ectopic NCC condensation at the level of the maxillary bud could evolve in cartilaginous fragmented elements (if the contribution of NCC-derived ectomesenchyme is limited) or in a skeletal extra element quite similar to the mandibular Meckel's cartilage (if a conspicuous NCC migration in the maxillary part of the branchial arch has taken place). This hypothesis is also supported by the observation that the ectopic maxillary cartilage is usually fused with cartilaginous elements at the base of the skull, suggesting a common anatomic origin of the two structures.

The abnormal migration of NCC in FON groups is also documented by the abnormal organization of the cranial nerves (cranial nerves and ganglia have a double origin: from NCC-derived ectomesenchyme and from placodes) (Whitlock, 2004).

The maxillary ectopic cartilage has been described by Morriss-Kay (1993) after treatment of pregnant mice with teratogenic dosages of RA as a displacement of Meckel's cartilage from the mandibular to the maxillary region. The data obtained in the present work suggest a different explanation to justify the ectopic maxillary cartilage: the abnormal migration of part of NCC coded to condense at the mandibular level to form cartilage gives rise to ectopic ectomesenchyme that differentiates and forms the ectopic cartilaginous bar.

In conclusion, this study shows that: 1) FON is teratogenic in rats; 2) FON induces abnormalities similar to those observed after RA treatment, including the formation of the maxillary cartilage; and 3) the maxillary cartilage originates by differentiation of the ectomesenchyme condensed from the ectopically-migrated encephalic NCC.

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